PAVAN DHARMA ADAPA

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PROFESSIONAL SUMMARY

Healthcare Data Analyst with an M.S. in Computer Science & Engineering, **available now for full-time roles**. Designs production ETL pipelines, dimensional data warehouses, and clinical-operations reporting across SQL, Python, Tableau, and Power BI. Portfolio work runs on **real public health data** — CDC NHAMCS emergency department microdata, CMS claims-denial filings, and multi-hospital inpatient encounters — with design-correct survey inference, calibrated risk models, and terminology-driven quality measures. Currently building and validating a genomics data pipeline for a Mississippi State research lab. Grounded in EHR/EMR concepts, HL7, ICD-10/CPT, SNOMED CT, LOINC, and HIPAA, and experienced translating clinical questions into validated data models and provider-facing reports mirroring Epic Clarity, Cogito, and Caboodle workflows.

TECHNICAL SKILLS

Epic Ecosystem & Reporting (self-study / certification prep): Clarity data model (ENCOUNTER, PAT_ENC, ORDER_PROC, RESULT, PAT_DIAGNOSIS), Cogito, Caboodle, Reporting Workbench concepts, SlicerDicer, Radar dashboards; Crystal Reports, SSRS, SAP BusinessObjects; Epic Sphinx assessment prep.

Healthcare Data & Standards: EHR/EMR systems, clinical & claims data, revenue cycle (denials, appeals, A/R, charge capture), population health, patient registries, quality measures (HEDIS, CMS eCQM), HL7 v2, ICD-10/CPT, SNOMED CT, LOINC, HIPAA compliance.

Data Engineering & Governance: ETL/ELT pipeline development, data warehousing, dimensional / star-schema modeling, grain design, data validation, cleansing & standardization, data quality, data governance.

Programming & Querying: SQL (joins, CTEs, window functions, query optimization), Python (Pandas, NumPy, scikit-learn, SciPy), PostgreSQL, MySQL, SQLite, Advanced Excel.

BI & Visualization: Tableau, Microsoft Power BI, Matplotlib, Seaborn — interactive dashboards, KPI reporting, executive-ready visualization.

Analytics & Modeling: Complex-survey estimation (weighting, ultimate-cluster variance, Taylor linearisation), hypothesis testing, EDA, risk stratification, predictive modeling (logistic regression, Random Forest, XGBoost, gradient boosting), model calibration, decision curve analysis.

Engineering Practice: Git, GitHub Actions CI, pytest, Snakemake, conda/Bioconda, Linux (WSL2), reproducible pipelines, hermetic test suites, code review, technical documentation.

Tools & Cloud: Linux, AWS, Jupyter, Selenium, SQLite.

PROFESSIONAL EXPERIENCE

Data Analytics Intern — Bioinformatics Pipeline — Jul 2026 – Present

Agricultural Microbiomes Lab, Mississippi State University — Starkville, MS

- Built and validated the pipeline the lab uses to turn Oxford Nanopore DNA sequencing output into bacterial abundance tables for poultry and catfish microbiome research.
- Ported the pipeline from the university's SLURM/HPC cluster to standard Windows and Linux workstations, removing every scheduler and cluster-storage dependency and cutting setup from a multi-day process to a single documented install.
- Cross-validated output against an independently developed implementation: the two agreed to **within 0.01 percentage points** on every species and genus, with identical read counts — the check that made the lab's results defensible.
- Processed and quality-controlled **2M+ sequencing reads** across 19 samples, and automated construction of the reference database from NCBI 16S RefSeq (27,700+ sequences, 21,000 taxa) into a single command.
- Cut a taxonomy-lookup step's memory requirement from **32 GB to under 1 GB** by replacing a full in-memory load with a streaming filter, and removed a 9 GB external data dependency.
- Root-caused an intermittent failure that destroyed results after an hour of compute, tracing it to host/guest clock drift rather than the cause the error message named; delivered the fix and the documentation.
- Contributed **6 merged pull requests** to the lab's open-source tool ahead of public release, covering packaging, failure diagnostics, citation metadata and contributor tooling.

Graduate Teaching Assistant & Analytics Mentor — Sep 2025 – May 2026

Mississippi State University — Starkville, MS

- Trained and supported **100+ students** across 6 courses on data cleaning, transformation, visualization, and dashboard-based reporting in SQL, Python (Pandas, Matplotlib), and Advanced Excel.
- Enabled early identification of at-risk students by tracking KPIs (completion rates, performance distribution, retention) in SQL and Excel, improving intervention turnaround.
- Created reusable training materials and documentation that improved task-completion speed by **25%** and standardized analytics best practices across cohorts.

Graduate Research Analyst — Healthcare Data Analytics — Aug 2024 – Aug 2025

Mississippi State University — Starkville, MS

- Designed and automated a production ETL pipeline (Python, Selenium, SQL) to extract, cleanse, and standardize multi-source clinical and surveillance datasets, cutting data-processing time **95%** — from several hours to under 10 minutes per dataset.

- Partnered with cross-functional stakeholders to translate clinical and operational questions into technical requirements, querying **900,000+** public-health surveillance records in SQL and Python to surface **8 key drivers** of disease transmission.
- Developed risk-stratification and predictive models (logistic regression, Random Forest, XGBoost) reaching **87% forecast accuracy**, and established baseline performance KPIs to monitor model and data quality over time.
- Reduced manual analysis time **50%** and resolved **35%** of dataset inconsistencies by automating feature-engineering and data-cleansing routines.

HEALTHCARE ANALYTICS PROJECTS

Interactive Analytics Dashboard · Five-view BI application over every project's output

- Built a five-view analytics dashboard — clinical operations, revenue cycle, population health, risk model, experimentation — with **14 interactive charts, 21 KPI cards and drill-down tables**, covering every project below.
- Data layer is generated directly from each pipeline's output files rather than re-entered, so the dashboard **cannot drift** from the analyses it reports; charts are hand-written SVG with no third-party dependency.

Emergency Department Throughput Analytics · CDC/NCHS NHAMCS — real national survey microdata · github.com/adapapavandharma/ed-throughput-analytics

- Analyzed **16,025 real emergency department visits**, survey-weighted to the **155.4M** US ED visits of 2022; the pipeline self-validates by reproducing CDC's published national total exactly or refusing to run.
- Implemented the **ultimate-cluster variance estimator with Taylor linearisation** CDC prescribes for this stratified multi-stage sample; design-correct inference showed the hypothesised peak-hour effect does **not** clear significance at $\alpha = 0.01$ — a negative result reported rather than discarded.
- Found between-hospital variation in median door-to-provider time (2–74 min) to be **3.9× larger** than time-of-day variation, and quantified boarding at **48.6M ED bay-hours lost annually** as the recoverable capacity item.

30-Day Readmission Risk & Care-Management Registry · 99,343 real inpatient encounters, 130 hospitals · github.com/adapapavandharma/readmission-risk

- Built a gradient-boosted readmission model (**AUC 0.664**) on real multi-hospital encounter data; top-quintile targeting captures **37.7% of all readmissions** at **1.89× enrichment**, number-needed-to-screen 4.8, delivered as a capacity-derived tiered risk board.
- Benchmarked against a single-variable clinical rule to isolate marginal value — prior inpatient count alone reaches 32.2% recall, so the model buys **+5.5 points**, materially less than an AUC comparison implies.
- Showed balanced class weighting inflates mean predicted risk to **46% against an observed 11%** while leaving AUC unchanged, establishing calibration rather than discrimination as the constraint on deployability.

Claims Denials & Appeals Analytics · CMS Transparency in Coverage — 3-year national panel · github.com/adapapavandharma/denials-analytics

- Built a 15,545 plan-year panel covering **126.9 million denied claims** across every Qualified Health Plan on the federal Marketplace, 2021–2023.
- Established that only **0.168% of denials are ever appealed** while **40.1% of appeals succeed**, identifying capacity to file — not claim merit — as the binding constraint on denial recovery.
- Showed the payer-reported reason taxonomy cannot support root-cause work (**"Other" is 71.7%** of categorised denials and growing), and that denial rate is an organisational property: **6.0% to 48.3%** across issuers with 100,000+ claims.
- Modelled two fact grains because CMS repeats issuer-level fields across every plan an issuer sells, and classified suppression markers before coercion — a value withheld for a small cell is missing-not-at-random, so averaging the survivors biases every aggregate upward.

Clinical Quality Measures Engine · HEDIS / CMS eCQM over a 13,810-patient EHR · github.com/adapapavandharma/clinical-quality-measures

- Built a declarative measure engine evaluating HEDIS and CMS eCQM specifications over **820,993 encounters and 10.8M observations**; measures are versioned YAML specs with SNOMED and LOINC value sets, so an annual steward update is a reviewable diff rather than a code change.
- Decomposed CMS122 to show its 17.3% poor-control score comprised **1 patient failing on a recorded result and 203 never tested**, reclassifying the finding from a glycaemic-control problem to a testing-outreach problem.
- Exposed a **32-point gap** in blood pressure control between ages 18–44 (57%) and 75+ (89%) hidden by the 66.9% aggregate, and delivered a care-gap registry of 4,439 gaps prioritising the **915 patients with two or more** closable in one contact.

EDUCATION

M.S., Computer Science & Engineering Graduated May 2026

Mississippi State University — Mississippi State, MS

Coursework: Advanced Machine Learning, Artificial Intelligence, Algorithms, Parallel Algorithms, Software Engineering, Data Analytics, Advanced Cyber Operations.

B.Tech, Computer Science & Engineering Graduated May 2024

Malla Reddy University — Hyderabad, India

PUBLICATION (IN PROGRESS)

Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production

Zhang Li, Pavan Dharma Adapa, Sai Harika Gade, Saikanth Ratnavale, Navicky Michael — compartmental (SIR/SEIR) and machine-learning forecasting of respiratory disease transmission at scale.